Leakage Risk and Class-Level Errors in CNN Classification of COVID-19 and Pneumonia on a Public Chest X-ray Dataset

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***Abstract*—Chest X-ray reading is hard when diseases look alike. We compared four convolutional neural networks on a public dataset of 5,073 images. The classes were normal, COVID-19, viral pneumonia, and bacterial pneumonia. The networks were a custom CNN, AlexNet, VGG16, and ResNet50. One held out set of 761 images tested every model. ResNet50 had the highest point estimates. Its accuracy was 83.4% (95% CI 80.6 to 85.9) and its macro F1 score was 83.3%. The other three models reached 78.3% to 80.0% accuracy. All four intervals overlap, so the ranking is tentative. Viral and bacterial pneumonia caused most errors. Viral recall stayed below 61% for every model. VGG16 also scored about 15 points higher on validation data than on test data. We then audited the folder structure described on the dataset page. It points to exact copies and augmented variants of training images inside the pool. If the page is accurate, a random image split leaves about two thirds of test images with a twin in the development data. Test scores may therefore be inflated. Patient level splits, deduplication, repeated runs, and external validation are needed before any clinical claim.**

***Index Terms*—Chest X-ray, COVID-19, pneumonia, convolutional neural network, data leakage, Grad-CAM, medical image analysis.**

I. Introduction

COVID-19 is caused by the SARS-CoV-2 virus. The World Health Organization declared a pandemic on 11 March 2020 [1]. Early patients often had pneumonia. In one cohort of 41 patients, every patient had abnormal findings on chest CT [2]. Fast triage became a priority for crowded health services.

Real time RT-PCR is the standard laboratory test [3]. It needs laboratory capacity and takes time. Sensitivity at first presentation can also be limited. One series of 51 patients found a positive first RT-PCR result in 71% of cases [4]. These limits keep interest alive in complementary tools.

Chest X-ray is cheap and widely available. Reading it is still difficult. Viral pneumonia, bacterial pneumonia, and COVID-19 can produce similar opacities. Image quality also changes with the device, the patient position, and the exposure.

Deep learning is now common in medical image analysis [5]. A 121 layer network reached radiologist level on pneumonia detection in a large chest X-ray collection [6]. Many groups then applied CNNs to COVID-19 radiographs. Early studies reported accuracies near 98% on small collections [9], [11]. Later work moved to larger benchmarks [12], [13].

Two problems weaken many of these scores. The first is leakage. Related images can sit in both the training and test partitions. The second is shortcut learning. A network can learn image borders, text markers, or the source of the image instead of disease [18]–[20]. A systematic review found major flaws and bias in the COVID-19 imaging models it examined [16]. Another review found that few public datasets allowed a proper bias assessment [17].

This study works on a public dataset that merges several sources. We compare four CNNs on one task and one test set. We report class level results, confidence intervals, and chance adjusted agreement. We also read the dataset description closely and ask what it implies for leakage.

Our contributions are as follows.

1) A controlled comparison of a custom CNN, AlexNet, VGG16, and ResNet50 on one shared test set, with full class level reporting.
2) An audit of the dataset structure based on the repository description. It shows that the study pool probably holds exact copies and augmented variants, and it estimates how many test images are exposed.
3) A documented gap between validation and test accuracy, with a list of candidate causes.
4) An exploratory Grad-CAM review that flags attention to borders and markers in some cases.

The paper is organised as follows. Section II reviews related work. Section III describes the data and methods. Section IV gives the results. Section V discusses them and states the limits. Section VI concludes.

II. Related Work

A. Deep Learning and Transfer Learning in Medical Imaging

Convolutional networks learn local patterns in early layers. Deeper layers combine them into larger structures. Litjens et al. surveyed more than 300 contributions and showed how widely the method had spread [5]. Rajpurkar et al. trained CheXNet on ChestX-ray14 and reported F1 performance above the average of four radiologists [6].

Medical datasets are usually small, so transfer learning is common. Yosinski et al. showed that first layer features are general and transfer well, while later layers become task specific [7]. Raghu et al. tested transfer on two large medical

tasks, one of them chest X-ray. They found little benefit from ImageNet weights. Small custom networks performed comparably to large pretrained models [8]. This result matters for the present comparison.

B. COVID-19 Detection from Chest X-ray

Apostolopoulos and Mpesiana used transfer learning on 1,427 images and reported 97.82% accuracy for COVID-19 detection [9]. Narin et al. tested pretrained networks on small public collections [10]. Ozturk et al. proposed DarkCovidNet and reported 98.08% accuracy for two classes and 87.02% for three classes [11]. Wang et al. released COVID-Net with the COVIDx benchmark of 13,975 images from 13,870 patients. COVID-Net reached 93.3% test accuracy [12].

Later work compared many architectures. Breve tested 21 CNNs on COVIDx8B, which holds 16,352 images. Each model was trained five times. DenseNet169 gave the best single model accuracy of 98.15%. A five model ensemble reached 99.25% [13]. Banerjee et al. blended snapshots of one CNN in an ensemble [14]. Li et al. added a contrastive learning task to improve generalisation on CT and X-ray data [15].

C. Dataset Bias, Leakage, and Shortcut Learning

Roberts et al. reviewed COVID-19 imaging models from 2020. They found major flaws and bias that made the models unsuitable for clinical use [16]. Some training sets used children as non COVID controls and adults as COVID cases. A model can then learn age instead of disease. Garcia Santa Cruz et al. reviewed public COVID-19 X-ray datasets and reached a similar warning [17].

Several studies give direct evidence of shortcuts. Maguolo and Nanni blacked out the lung region and trained only on the outer image. Results stayed similar, so the networks used cues unrelated to the lungs [18]. DeGrave et al. used explainable methods and found that COVID-19 models relied on confounders. The models looked accurate but failed in new hospitals [20]. Zech et al. found that CNNs could identify the hospital system of a radiograph with near perfect accuracy. Performance on outside hospitals was lower in three of five comparisons [19]. Geirhos et al. named this behaviour shortcut learning [21].

D. Research Gap

Three gaps motivate this study. Few papers compare networks of different design under one protocol and one test set. Overall accuracy is often reported without class level detail. Dataset structure is often accepted without a check for duplicates. We do not claim to close these gaps. We offer a transparent comparison and a clear account of what the data can support.

III. Materials and Methods

A. Dataset and Structural Audit

We used the COVID-19 Detection X-Ray Dataset on Kaggle [22]. The page describes X-ray images of COVID-19, bacterial pneumonia, viral pneumonia, and normal cases. It names two sources. The COVID-19 images come from the collection of Cohen et al. [23]. That collection was assembled from publications and websites. The pneumonia and normal images come from a Kaggle pneumonia dataset based on Kermany et al. [24]. Those images are from patients aged one to five years at one children's hospital in Guangzhou. The Kaggle page calls the images posteroanterior views. The Kermany dataset description gives anterior posterior views. This mismatch is one more sign of mixed provenance.

TABLE I: Folder counts listed on the dataset page and the implied merged pool

Class	Train	Val	Non aug. (assumed)	Implied pool	Study share (%)
Normal	901	450	901	2,252	44.0
COVID-19	120	9	60	189	3.7
Viral	412	205	412	1,029	20.3
Bacterial	650	324	650	1,624	32.0
Total	2,083	988	2,023	5,094	100.0

Counts are taken from the dataset page. The non augmented column assumes that the 60 augmented COVID-19 images are absent.

The page lists three folders. TrainData holds 2,083 images. ValData holds 988 images. NonAugmentedTrain is described as the same data as TrainData without augmentation. TrainData lists 60 original and 60 augmented COVID-19 images. Table I shows the counts. It also shows the pool we obtain if all three folders are merged and the third folder holds the original images of the first.

The study pool held 5,073 images. The implied pool holds 5,094. The class shares of the study pool match the implied counts for COVID-19, viral, and bacterial images. Normal images differ by about 20. We cannot explain this small gap from the files alone. The match still suggests that all three folders were merged.

Two consequences follow. First, if the folder descriptions are accurate, the pool holds at most 3,011 distinct source images. About 2,083 pool images, or 41%, are copies or augmented variants. Second, a random image split then places many twins on both sides. Our test set had 761 images, or 15.0% of the pool. Under a random split, a pooled image with one twin has a 0.85 chance that the twin lies in the development data. A COVID-19 image from a group of three has a 0.98 chance that at least one twin does. Weighting by group size gives about 69% of test images with a twin in the development data. For COVID-19 test images the figure is about 93%. These are expected values under stated assumptions. They are not measured counts. A hash based duplicate check is needed to confirm them.

B. Data Partition

A test set of 761 images was held out for final evaluation. It contains 29 COVID-19, 334 normal, 155 viral pneumonia, and 243 bacterial pneumonia images (Table II). The split was made by image with a fixed random seed of [seed value]. No deduplication and no patient grouping were applied. The remaining images were divided into training and validation

TABLE II: Composition of the held out test set

Class	Images	Share (%)
COVID-19	29	3.8
Normal	334	43.9
Viral pneumonia	155	20.4
Bacterial pneumonia	243	31.9
Total	761	100.0

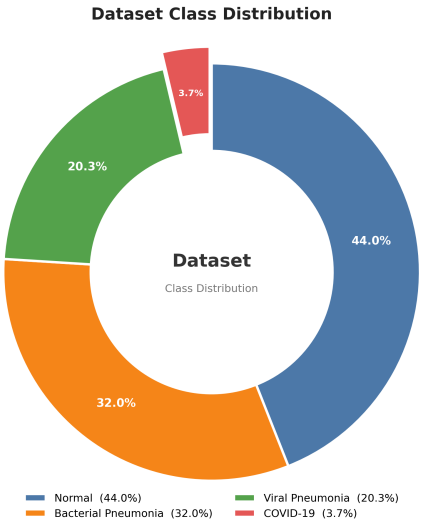


Fig. 1: Class distribution of the 5,073 image pool.

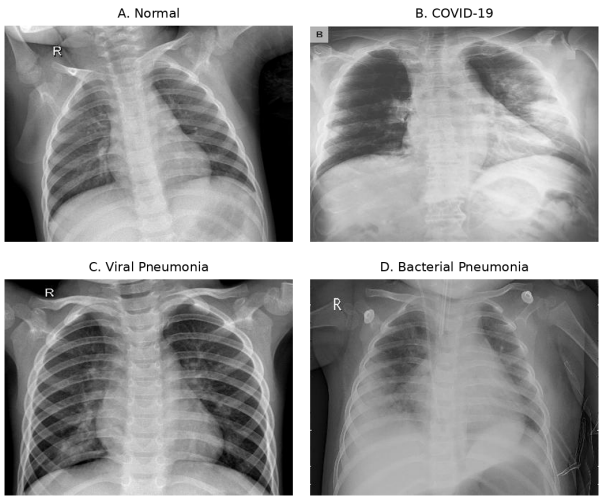


Fig. 2: Representative chest X-ray images from the four classes.

subsets at a ratio of [train to validation ratio]. The test set was not used for training or model selection. Every network was scored on the same test images.

C. Preprocessing and Augmentation

Images were resized to 224×224 pixels. AlexNet used 227×227 input, as shown in its architecture diagram. Pixel values were rescaled to the range zero to one. Histogram equalisation improved contrast. Data were then normalised.

TABLE III: Summary of the evaluated networks

Model	Initialisation	Input	Parameters (M)
Custom CNN	Random	224×224	25.8
AlexNet	Random	227×227	58.3
VGG16	ImageNet	224×224	134.3
ResNet50	ImageNet	224×224	27.8

Parameter counts were computed from the layer settings in Fig. 3.

Random augmentation was applied to training batches only. It included horizontal flips, rotation, zoom, translation, and brightness changes. The repository folders were not filtered for augmented files. Augmented images may therefore also appear in the test set. The COVID-19 example in Fig. 7 shows rotation and fill artefacts that look like repository augmentation.

The classes are unbalanced. We used class weights in the loss, so rare classes received larger weights [25].

D. Network Architectures

We compared four networks (Table III). The custom CNN and AlexNet [26] started from random weights. VGG16 [27] and ResNet50 [28] started from ImageNet weights. The comparison therefore covers whole training recipes. It cannot isolate architecture.

The custom CNN has three convolutional blocks with 32, 64, and 128 filters. Each block uses a 3×3 convolution, batch normalisation, max pooling, and spatial dropout. Dense layers of 256 and 128 units follow. A softmax layer outputs four probabilities. AlexNet has five convolutional layers and two dense layers of 4,096 units with dropout of 0.5. VGG16 has 13 convolutional layers and two dense layers of 4,096 units. ResNet50 uses bottleneck residual blocks, global average pooling, and one dense layer of 2,048 units. Fig. 3 shows all four designs.

E. Training Procedure

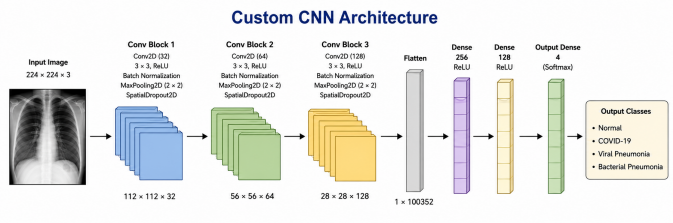
All models used the Adam optimiser [29] with an initial learning rate of 10^{-4}. The loss was categorical cross entropy and the batch size was 16. Training ran for at most 60 epochs with early stopping [30]. Table IV lists the settings.

VGG16 and ResNet50 were trained in two stages. In stage one, the backbone was frozen and the head trained for 30 epochs. In stage two, upper backbone layers were unfrozen and tuned for up to 30 more epochs. For VGG16 the upper eight layers were unfrozen. For ResNet50 the number was [number of layers]. Early stopping ended both pretrained runs at about 42 to 43 epochs in total. The custom CNN and AlexNet ran for 60 epochs.

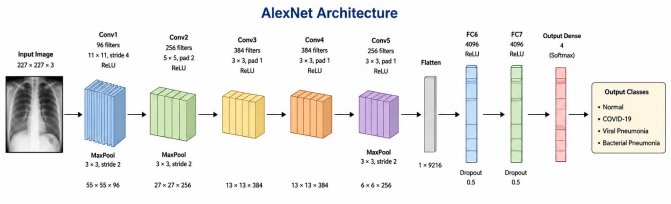
Training used one NVIDIA RTX 3060 GPU with 12 GB of memory. The code used TensorFlow 2 and Keras.

F. Evaluation Metrics

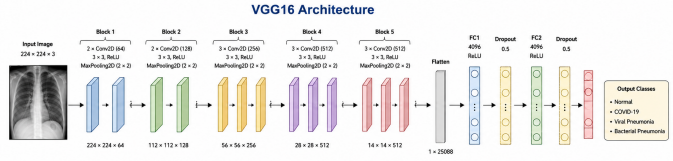
We scored every model on the held out test set. We report accuracy, per class precision, recall, and F1, and macro averages. Macro averages give each class equal weight. They are stricter than accuracy on unbalanced data.



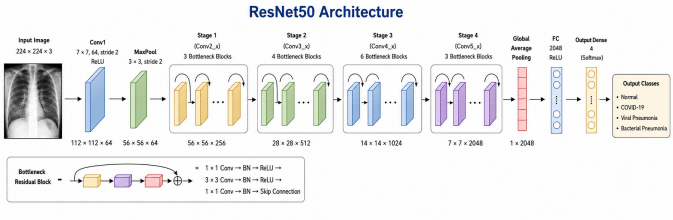
(a) Custom CNN



(b) AlexNet



(c) VGG16



(d) ResNet50

Fig. 3: Architectures of the four evaluated networks.

TABLE IV: Training settings

Setting	Value
Optimiser	Adam
Initial learning rate	10^{-4}
Loss	Categorical cross entropy
Class weighting	Yes
Batch size	16
Maximum epochs	60
Early stopping patience	[patience]
Fine tuned layers, VGG16	Upper 8
Fine tuned layers, ResNet50	[number of layers]
Random seed	[seed value]
Runs per model	1
Train to validation ratio	[ratio]

We added two chance adjusted measures. Cohen's kappa compares observed agreement with agreement expected by chance. The multiclass Matthews correlation coefficient (MCC) summarises the full confusion matrix in one value [31]. We computed 95% confidence intervals for accuracy and recall with the Wilson score method [32]. This method treats images as independent. Twins in the data break that assumption, so the intervals are probably too narrow. All values were computed from the test confusion matrices.

G. Interpretability Analysis

We inspected first layer feature maps to see what early filters respond to. We used Grad-CAM [33] to mark the regions that most affected a prediction. The maps in this paper come from [model name] at layer [layer name]. Both tools are qualitative. We do not treat them as proof of clinically valid features.

IV. Results

A. Overall Test Performance

Table V gives the overall results. ResNet50 had the highest value on every aggregate metric. Its accuracy was 83.4%. The custom CNN and AlexNet were close at 79.9% and 80.0%. VGG16 was lowest at 78.3%.

ResNet50 led the custom CNN by 3.6 points, AlexNet by 3.4 points, and VGG16 by 5.1 points. All four confidence intervals overlap, which is a cautious sign but not a formal test. We lack paired predictions, so we ran an unpaired two proportion test. It ignores pairing and is conservative when the models err on the same images. It gave $p = 0.07$ against the custom CNN, $p = 0.08$ against AlexNet, and $p = 0.01$ against VGG16. Only the VGG16 gap clears the usual 0.05 level, and the test assumes independent images. The ranking is therefore indicative only. The lead is clearer in macro F1. ResNet50 scored 83.3%, against 76.8%, 76.5%, and 73.7% for the others.

B. Per Class Results and Error Structure

Table VI lists recall, precision, and F1 for each class. Fig. 4 shows the confusion matrices.

Normal images were easiest. Recall ranged from 95.2% to 99.4%. COVID-19 results varied. ResNet50 found 28 of 29 cases, a recall of 96.6% (95% CI 82.8 to 99.4). AlexNet found 20 of 29, a recall of 69.0%. Intervals for this class are wide because only 29 images were available.

Viral pneumonia was hardest. Recall ranged from 38.1% for VGG16 to 60.6% for ResNet50. Most errors lay between the two pneumonia types. VGG16 labelled 75 of 155 viral images as bacterial. ResNet50 did so for 51. The reverse error was also common. ResNet50 labelled 54 of 243 bacterial images as viral. Separating viral from bacterial pneumonia is the weak point of all four networks. Overall accuracy hides this.

C. Training Behaviour and the Validation to Test Gap

Fig. 5 shows training and validation accuracy. Training accuracy rose steadily for all four networks. VGG16 reached about 0.99. Its validation accuracy ended near 0.93. The other final validation values were about 0.89 for ResNet50, 0.88 for

TABLE V: Test performance of the four networks on 761 held out images

Model	Accuracy (%)	95% CI	Macro P (%)	Macro R (%)	Macro F1 (%)	Kappa / MCC
Custom CNN	79.89	76.9 to 82.6	77.21	76.53	76.75	0.694 / 0.695
AlexNet	80.03	77.0 to 82.7	80.01	74.01	76.47	0.697 / 0.697
VGG16	78.32	75.3 to 81.1	75.72	73.03	73.72	0.666 / 0.670
ResNet50	**83.44**	**80.6 to 85.9**	**83.80**	**82.77**	**83.25**	**0.749 / 0.749**

P is precision and R is recall. The interval is the Wilson 95% interval for accuracy. All values come from the test confusion matrices.

TABLE VI: Per class recall, precision, and F1 (%) on the test set

Class (n)	Custom CNN R / P / F1	AlexNet R / P / F1	VGG16 R / P / F1	ResNet50 R / P / F1
COVID-19 (29)	82.8 / 82.8 / 82.8	69.0 / 95.2 / 80.0	79.3 / 85.2 / 82.1	96.6 / 100.0 / 98.2
Normal (334)	95.8 / 92.8 / 94.3	95.2 / 95.2 / 95.2	98.2 / 89.6 / 93.7	99.4 / 94.9 / 97.1
Viral (155)	52.3 / 61.8 / 56.6	56.1 / 56.9 / 56.5	38.1 / 58.4 / 46.1	60.6 / 62.7 / 61.6
Bacterial (243)	75.3 / 71.5 / 73.3	75.7 / 72.7 / 74.2	76.5 / 69.7 / 72.9	74.5 / 77.7 / 76.1

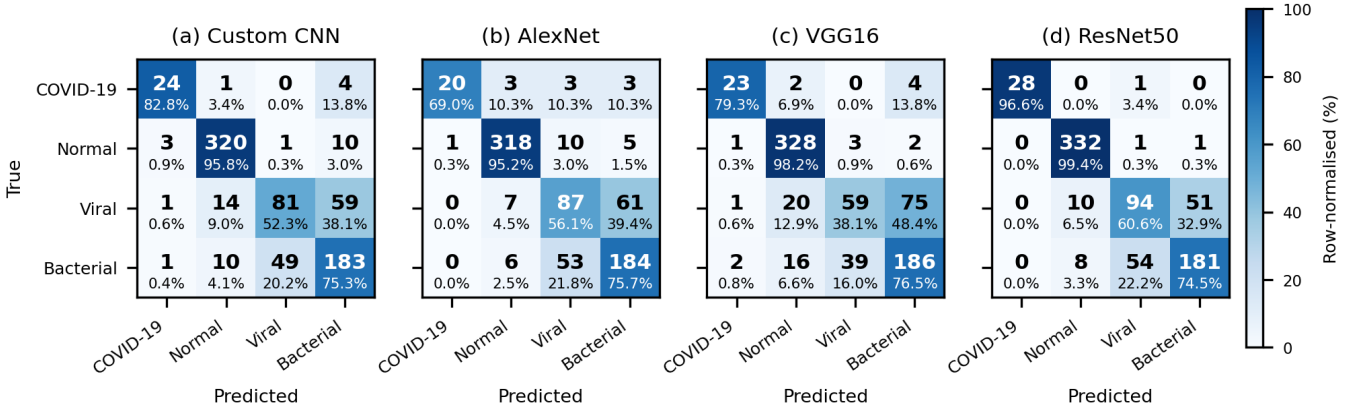


Fig. 4: Confusion matrices on the 761 image test set. Bold numbers are image counts. Small numbers are row percentages, which equal per class recall.

the custom CNN, and 0.80 for AlexNet. We read these values from the plot. Exact logged values should replace them.

Test accuracy was lower than validation accuracy for three models. The gap was about 15 points for VGG16, 6 for ResNet50, and 8 for the custom CNN. AlexNet showed almost none. Its validation curve was very unstable and fell below 0.1 in two epochs.

A gap of 15 points needs an explanation. We did not test any of the following causes.

1) Twins in the training and validation subsets could inflate validation accuracy. This follows from the structure in Table I.
2) Early stopping selects the best epoch on the validation set. This adds an optimistic bias.
3) The class mix of the validation subset may differ from that of the test set.
4) Folders in the repository differ in image shape. TrainData images are listed as 300×400, while ValData images have variable shapes. A shift between sources could matter.

A direct test of each cause needs saved predictions and a deduplicated split.

D. Feature Maps

Fig. 6 shows first layer activations for the custom CNN, VGG16, and ResNet50. Several filters respond to rib edges and to intensity changes at the lung boundary. Others pass a smoothed copy of the image. This is typical for early layers [7]. The first layer maps of AlexNet were almost flat and are not shown. These maps do not show that any filter encodes a disease specific finding.

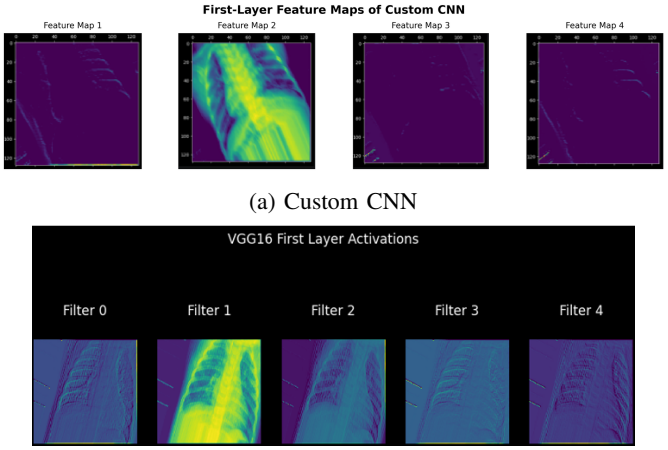
E. Grad-CAM Analysis

Fig. 7 shows Grad-CAM maps for four selected cases. In the viral pneumonia case, the strongest response lies inside the lung fields. This is where findings are expected.

Other maps are less reassuring. In the normal case, strong responses appear near the shoulder, the upper border, and the laterality marker. In the COVID-19 case, the image has rotation and fill artefacts, and the response spreads across the frame. Some predictions may therefore lean on non anatomical cues. Four cases cannot show how often this happens. We treat the maps as exploratory.

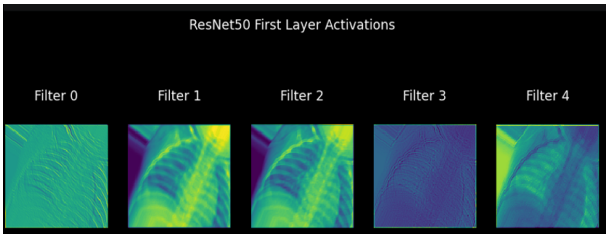


Fig. 5: Training (solid) and validation (dashed) accuracy for the four networks. VGG16 and ResNet50 are fine tuned models.



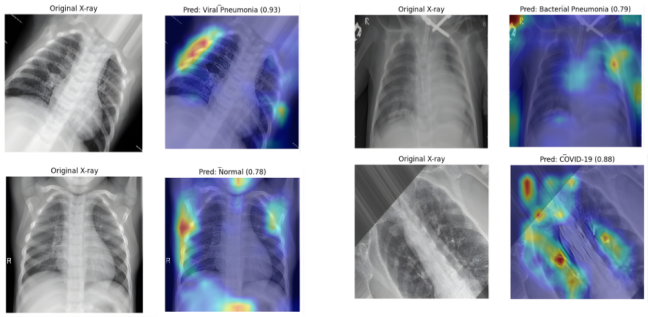
(a) Custom CNN

(b) VGG16



(c) ResNet50

Fig. 6: First layer feature maps for one input image.



(a) Viral pneumonia (top) and normal (bottom)

(b) Bacterial pneumonia (top) and COVID-19 (bottom)

Fig. 7: Grad-CAM maps for selected cases. Each row shows the input and the overlay with the predicted class and its probability.

V. Discussion

A. Principal Findings

ResNet50 gave the strongest and most balanced results. It had the best macro F1, the best COVID-19 recall, and the best viral pneumonia recall. VGG16 did worst, although it is pretrained and has the most parameters. The custom CNN and AlexNet start from random weights and scored close to each other.

Initialisation does not predict rank here. A pretrained model finished first and another pretrained model finished last. We trained no scratch version of either network. We also trained no pretrained version of the custom CNN or AlexNet. The design therefore cannot separate the effect of initialisation from the effect of architecture. The result fits the finding that ImageNet weights help medical tasks less reliably than often assumed [8]. The very large dense head of VGG16 may also overfit. We did not test this idea.

B. Why Accuracy Alone Misleads

Accuracy sat between 78% and 83% for all models. The range hides large differences between classes. Normal images were found almost perfectly. Viral pneumonia was found in fewer than 61% of cases by every model. A clinician would care most about missed cases and about confusion between

pneumonia types. Macro scores, kappa, and confusion matrices expose these weaknesses. A single accuracy value does not.

C. Dataset Construction, Leakage, and Shortcut Risk

The dataset merges sources that differ in age, setting, and processing. The pneumonia and normal images come from children aged one to five [24]. The COVID-19 images were collected from publications and websites [23]. A network can separate classes by source instead of by disease. DeGrave et al. and Maguolo and Nanni documented this problem [18], [20]. The Grad-CAM maps here show attention to borders and markers in some cases. This is consistent with a shortcut risk. It does not prove that shortcuts drove the predictions.

Leakage is the second risk, and our audit makes it concrete. The folder description implies exact copies and augmented variants. A random image split then places twins on both sides. We estimate that about 69% of test images, and about 93% of COVID-19 test images, have a twin in the development data. If so, a model can score well by recognising images it has seen. The high COVID-19 recall of ResNet50 should be read with this in mind. The observed accuracies of 78% to 83% sit below what full memorisation of twins would suggest. Either the overlap is smaller than estimated or the networks only partly memorised it. Augmented variants differ from their originals, which would also weaken the effect. We cannot tell these cases apart without a duplicate check. The Wilson intervals assume independent images, so they are probably too narrow. No patient identifiers exist, and a patient level split cannot be checked.

D. Relation to Published Work

Published scores are often higher than ours. COVID-Net reached 93.3% on a three class task with a larger benchmark [12]. Breve reported 98.15% for the best single DenseNet169 on COVIDx8B [13]. We cannot compare these numbers directly. The studies differ in data sources, class definitions, split rules, and repeated runs. Our task is also harder than a binary one, because it includes the viral and bacterial distinction. Breve's comparison also covered DenseNet and EfficientNet families, which we did not test. Our value lies in a transparent comparison and a dataset audit. It does not lie in a rank against other papers.

E. Practical Implications

These results do not support clinical use of any model. Viral pneumonia recall is too low. The COVID-19 class is too small for stable estimates. External validation is missing. The work can serve as a baseline on this dataset. Researchers who use it should deduplicate first. They should report class level results with intervals. They should also state how they split the data.

F. Limitations

The study has several limits, and we group them by theme.

The first group concerns evidence. All results come from one split and one run per model. Overlapping intervals suggest that the ranking could change with repeated runs. We had no per image predictions, so paired tests such as McNemar's test could not be run. Threshold free measures such as the area under the ROC curve are also missing. Only 29 COVID-19 images were in the test set, and one error moves its recall by 3.4 points.

The second group concerns the data. The leakage figures are expected values from the repository description, and we ran no hash based duplicate check. The 21 image gap between the implied pool and the study pool is unexplained. The dataset has no patient identifiers, so a patient level split could not be built. No external or prospective data were used.

The third group concerns design. Pretrained and scratch models differ in more than architecture. Horizontal flips can create anatomically unusual images, because the heart normally lies on the left. Grad-CAM was reviewed by eye on four cases, and a test against lung masks is still needed. The models are not evidence of clinical readiness.

VI. Conclusion

We compared four CNNs on one chest X-ray task with four classes. ResNet50 had the highest point estimates, with 83.4% accuracy and a macro F1 score of 83.3% on 761 held out images. The custom CNN, AlexNet, and VGG16 reached 79.9%, 80.0%, and 78.3% accuracy. The intervals overlap, so the ranking is tentative.

All models struggled with viral pneumonia. Most errors lay between viral and bacterial cases. VGG16 showed a large gap between validation and test accuracy. The repository structure suggests that exact copies and augmented variants sit in the pool. Test scores may therefore be inflated.

Future work should deduplicate by hash and split by patient. It should repeat runs over several seeds and add paired tests. It should train scratch and pretrained versions of the same architectures, add newer architectures, and test on external multicentre data. It should also measure Grad-CAM against lung masks. These steps are needed before automated chest X-ray classification can support clinical decisions.

Data and Code Availability

The dataset is public on Kaggle [22]. The test set of 761 images was drawn with a fixed random seed. The list of image identifiers for each partition and the training code will be released at [repository link].

Ethics Statement

This study used public and de-identified chest X-ray images. No new patient data were collected and no human participants took part. Consent procedures rest with the original dataset providers.

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[Funding statement to be completed by the authors.] [Competing interests statement to be completed by the authors.]

Declaration of AI Assisted Writing

The authors used an AI language assistant to improve the wording and structure of this manuscript. The authors reviewed and edited all content. The authors take full responsibility for the work.

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